

Can Institutional Investors Reduce Security Issuance Frictions? Evidence from Anchor Backed IPOs in India.

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Abstract

We study the consequences of an exogenous regulation that allows IPO firms to recruit lead (anchor) investors in IPOs in India. We document a 7% reduction in underwriter compensation, a 26% reduction in issue discount, a 40% increase in capital investments and 12.3% reduction in financial leverage for anchor backed IPOs but not for non-anchor-backed firms. The effects on investments and leverage are significant for high growth and financially constrained firms. Anchor backed IPOs also exhibit better long-term operating performance. Our findings provide evidence that discretionary allocation of shares to institutional investors may reduce security issuance frictions for issuers.

Keywords: Issuance Frictions, Initial Public Offerings, Capital Structure, Investment

JEL Classification: G32, G18

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1. Introduction

The literature on Initial Public Offerings suggests that institutional investors in IPOs play several useful roles. There is considerable evidence on the importance of institutional investors in the pricing of initial public offerings (IPOs) and on the allocation to institutional investors. Rock (1986) finds that allocation to institutional investors mitigates the adverse selection problem faced by retail investors and ensure that retail investors do not withdraw from the IPO market. Benveniste and Spindt (1989) find that institutional investors help in providing value-relevant information necessary for the pricing of IPOs. Consistent with this view, Chemmanur (1993) finds that these informed investors help in bringing the market value of a firm's equity closer to its intrinsic value by their post-IPO trading activities. In a similar vein, Aggarwal, Prabhala and Puri (2002) provide empirical evidence that institutional allocations help the underwriters in extracting favorable pre-market demand information from them to adjust the offer price.

Allotting a large block of shares to institutional investors may be beneficial for retail investors as it serves as a signal of issue quality and mitigates information asymmetry *ex-ante*. Thus, allocation to institutional investors provides informational benefits to issuing firms and retail investors. When the degree of institutional ownership is high, it provides institutional investors the incentive to monitor a firm's performance. Therefore, IPOs backed by institutional investors may have superior long-term performance. Field and Lowry (2009) show that institutional ownership may have the ability to predict long-term risk-adjusted performance. Boehmer, Boehmer and Fishe (2006) show that institutional investors obtain more allocations in IPOs with better long-term performance.

In this paper we examine the role of institutional investors in reducing equity issuance frictions for firms that go public. We exploit a legal experiment resulting from the 2009 Securities

Exchange Board of India (SEBI) regulation permitting IPO managers to allocate shares on a preferential basis to anchor investors (AI, who are lead investors in the IPO) in the IPO pre-market, subject to disclosures relating to the price at which allocations were made and the identity of anchor investors. Specifically, we try to investigate the impact of anchor-backing on firms' issue discount, investment, and capital structure after the IPO and on investor demand and underwriter compensation in the pre-market. While institutional ownership may be useful for all types of firms because of their ability to participate in corporate governance, it may be particularly important for small and young firms that are going public. Our study is motivated by Seth, et. al; (2019) who document that anchor-backed IPOs earn superior returns through monitoring when anchor investors invest along with institutional investors who are known to be active monitors.

Our paper is related to Gustafson and Iliev (2017) who study the consequences of an exogenous deregulation in 2008 that allows smaller firms with a public float of less than \$75 m to raise capital via shelf registration. They document a reduction in equity issuance costs, an increase in investment, and a decline in financial leverage. In contrast, we use the aforementioned natural experiment to study how institutional investors can help IPO firms in achieving the same result in an emerging market context. Using a sample of Indian IPOs from 2006 to 2019, we provide evidence on the impact of anchor-backing on firms' investment and leverage, especially for high growth and financially constrained firms. We also study the long-term performance of these IPO firms. Our research design is broadly similar to that of Gustafson and Iliev (2017).

A distinguishing feature of the Indian book building mechanism is that underwriters are required to formally reveal their books to the general public unlike other countries. This enables the investors to view the aggregate demand on a real-time basis at different price points, which helps them in observing the extent of over-or-under subscription in each investor category. The

underwriters are required to allocate a predetermined fraction of shares for three categories of investors i.e. Qualified Institutional Buyers (QIBs), Non-Institutional Buyers (NIBs) and Retail Individual Buyers (RIBs). The SEBI July 2009 regulation introduced the fourth category of investors i.e. “anchor investors” who are allowed to participate in the pre-IPO market. The regulation stipulates that if the price fixed for public issue through book building is higher than the price at which anchor investors are allotted, they are required to pay the additional amount. However, if the price fixed for the public issue is lower than the price at which the allocation is made to anchor investors, then they are not paid the difference in price. The law also prohibits them from selling their allocations for 30 days after the IPO. This early participation by anchor investors provides a signal of issue quality and might be beneficial for issuers and investors.

Our analysis uses 788 IPOs that were issued between the first quarter of 2006 and the last quarter of 2019 out of which 154 were anchor backed. Our focus is on the impact of anchor investors in the pre- and after-market. To do this, we employ difference-in-difference (DiD) estimation in our analyses throughout the paper. We formulate a time window to segregate the pre- and post-legislation years around the introduction of the regulation in 2009. We exploit the regulatory reform to investigate whether the introduction of anchor investors reduce issuance frictions in the pre-market and a significant impact on financial policies.

Lowry, Officer, and Schwert (2010) raise concerns about the efficacy of the traditional book building mechanism because of underwriters’ difficulty to value firms characterized by higher information asymmetry. This paper sheds light on the efficacy of the regulatory reform in mitigating issue discount (underpricing). More specifically, we examine whether the allocation to anchor investors have an impact on IPO underpricing although we study other types of issuance frictions and the investment and leverage choices of firms.

Allotting a large block of shares to institutional investors results in economics of scale for the issuer and underwriter and may result in higher firm value due to monitoring by institutional investors (Shleifer and Vishny, 1986). It is likely that firms facing asymmetric information enlist anchor investors to ensure issue quality (Seth et al., 2019). We posit that anchor investor participation results in a reduction in security issuance frictions arising out of asymmetric information and agency problems and a higher investment and lower financial leverage particularly for growth firms and financially constrained firms.

We find that anchor backed IPOs have lower institutional subscription. Firms are more likely to recruit anchor investors when other institutional investors are not likely to invest thereby filling a void. The underwriter compensation falls by 7% when reputed anchor investors are involved and anchor backed IPOs have 26% lower underpricing, a 40% increase in capital investments and 12.3% reduction in financial leverage. The effects are economically, significantly, large for high growth and capital constrained firms. Financially constrained firms backed by anchor investors raise 26.4% more equity capital than unconstrained firms.

The rest of the paper is organized as follows. In section 2 we present a review of the related literature. Section 3 describes the institutional setting, data sources, sample, and summary statistics. In section 4 we present the empirical results. We provide additional analysis in Section 5. Section 6 concludes.

2. Related Literature

One of the potential benefits of the anchor investors program is that it reduces the risk of devolvement for underwriters. The “reduction in underwriting risk” hypothesis suggests that by pre-selling a large fraction of shares to anchor investors, underwriters can avoid the risk of under-

subscription. Since the process results in a reduction in underwriting risk we expect the underwriter compensation to reduce when anchor investors are involved even for riskier firms.

Seth et al., (2019) show that anchor backed IPOs have lower underpricing as compared to non-anchor backed IPOs. They study bidding by anchor investors in the two-stage initial public offering (IPO) process in India and document a decline in underpricing for anchor backed IPOs. They find that anchor investors are likely to invest in hard-to-place offerings characterized by valuation uncertainty and document a positive relationship between allocation to reputed anchor investors and returns up to lock-up expiration. Their study provides empirical evidence to support information revelation and targeting specific investors' theories of book building. Bhattacharya, et al., (2020) study anchor investors in 182 Indian IPOs during 2009–2017. They find that the share allotment and the number of anchor investors separately have significant impacts on valuation and underpricing; however, the net effect is insignificant. Further, anchor investors significantly influence other institutional investors' participation in the IPO and lower the aftermarket volatility. Samdani (2017) analyzes anchor backed IPOs in India and shows that anchor investor participation in IPOs reduces the heterogeneity of investor beliefs associated with accounting information, thereby improving financial reporting efficiency. We contribute to this literature by examining the role of anchor investors in reducing security issuance frictions particularly for growth and capital constrained firms.

Several empirical studies (Rock, 1986; Beatty and Ritter, 1986; Benveniste and Spindt, 1989; Chemmanur, 1993; Field and Lowry, 2009; Chemmanur et al., 2009) have analyzed the role of institutional investors in pricing and allocation of initial public offerings in the U.S. Field and Lowry (2009) find that IPO firms with higher institutional investment outperform firms with lower institutional investment. Investigating the pattern of institutional and non-institutional investment

for a large sample of IPOs issued in United States over 20 years, they document that institutional investors accurately use publicly available information to identify IPO firms that are likely to perform better. Aggarwal et al. (2011) find that changes in institutional ownership over time positively impact improvements in internal governance of firms. Demiralp et al., (2011) provide evidence on the monitoring benefits of institutional ownership by analyzing the impact of institutional ownership on stock price and operating performance following seasoned equity offerings. Using a sample of 3093 public equity issues completed during 1982–2006 in the U.S, they find that announcement returns are positively and significantly related to total and active institutional ownership levels and concentration. Further, they find a positive and significant relationship between post-issue stock returns and the contemporaneous post-issue changes in total and active institutional ownership and the concentration of their shareholdings. The study also shows that operating performance improvements are also related to institutional monitoring in the one, two, and three years following the equity issue. Their results support the notion that institutional investors have an informational advantage, which enables them to identify and invest in (subsequently) better performing firms. We contribute to this literature by whether anchor investor can improve the investee firms' ability to raise equity capital, reduce underwriter compensation, initial returns and leverage as well as increase investments.

Several authors have studied the design and impact of cornerstone investors (similar to anchor investors) in Asian markets particularly in Hong Kong (Esenlaub et al., 2016; McGuinness, 2012 and 2014; Tan and Ong, 2013). Esenlaub et al., (2016) find that IPOs backed by cornerstone investors with longer lock-up periods stay listed for a longer period. Thus, these authors support the monitoring role of institutional investors. We contribute to this literature by examining the effects of anchor investor participation on operating performance.

3. Institutional Setting, Data and Sample

3.1. Institutional Setting

In July 2009, SEBI introduced the concept of Anchor Investors (AIs) in the Indian capital market to boost investor confidence and also provide credible benchmarks for IPO valuation. The SEBI regulation requires AIs to be Qualified Investor Buyers⁴. The issuing firm can issue up to 30% of the allocation available to QIBs to anchor investors⁵. Anchor investors are required to invest a minimum of INR 100 million with a lock-in period of one month from the allotment date. The goal of this legislation is to attract reputable and sophisticated investors who could influence investors' demand by participating in the pre-IPO market. Further, one-third of the allocation available to anchor investors must be reserved for domestic mutual funds. The maximum number of anchor investors permitted in an IPO is 2 for offer size up to INR 100 million; a minimum of 5 and maximum of 25 anchor investors for offer sizes up to and beyond INR 2500 million respectively, subject to a minimum allotment of INR 50 million per investor. Typical anchor investors include commercial banks, mutual funds, foreign institutional investors, venture capital funds, insurance companies, provident funds and pension funds.

The SEBI regulation empowers the underwriters to identify anchor investors and determine an appropriate allotment price. Pricing and allocation (carved out from the QIB's tranche) must be finalised one day before the book opens to the general public. The regulation indicates that

⁴ Qualified institutional investors are institutional investors who are generally perceived to possess the expertise and the financial strength to evaluate and invest in securities and are to be registered with the Securities Exchange Board of India. There are around 300 QIBs in India.

⁵ The Indian securities law prescribes that (a) not less than 30% of the net offer be allotted to retail individual investors; (b) not less than 10% of the net offer be allotted to non-institutional investors i.e. investors other than retail individual investors and Qualified Institutional Buyers; and (c) not more than 60% of the net offer be allotted to Qualified Institutional Buyers. If QIBs apply for exactly the same number of shares earmarked for them, the offer would have a subscription of 1x in the QIB category. Thus, a QIB subscription of 10x implies an oversubscription of 9x in the QIB category.

there should not be any material relationship between the anchor investors and the book runner or the founders of the firm to ensure that anchor investors are independent. Another unique feature of the regulation is aimed at ensuring fair valuation. If the price quoted by anchor investors is higher than the offer price to the public, then the firm is not liable to refund the money to anchor investors. On the other hand, if the price fixed for anchor investors is less than the eventual offer price, then they are required to pay the difference to the firm. Once the price at which the allocation is made to the anchor investors is fixed, the regulation requires the underwriter to disclose the price and the quantum of shares allotted to the general public before the opening of the IPO to other investors. Since the investor knows the identity of the anchor investors and the price at which allocations are made to them, the remaining investors may use this as a relevant valuation benchmark.

3.2. Data and Sample

Our sample consists of 788 IPOs that were issued between 2006 and 2019 out of which 154 were backed by anchor investors. We obtain IPO related information from the Prime IPO Database and the website chittorgarh.com maintained exclusively for IPOs and supplement the data with historical stock prices from the Bombay Stock Exchange (BSE) website. We manually gather the data on allotment dates for each IPO from Business Line, a prominent business daily. We collect firm-level data from the Prowess database of the Centre for Monitoring Indian Economy, which is a standard source of information related to financial statements of Indian firms (Khanna and Palepu, 2000).

3.3. Variable Construction

Appendix A provides a summary of the key variables used in our analysis and the data sources.

We briefly discuss some of the important variables here.

Underpricing: We measure underpricing as the percentage return from the offer price to the first day closing price on the Bombay Stock Exchange. This is the definition used in prior papers (see, for e.g. Clarke et al., 2016).

Update: $((\text{Offer price} - \text{midpoint of the filing range}) / (\text{midpoint of the filing range})) \times 100\%$

Lead Manager Reputation: To measure the quality of the investment banker/underwriter we use Thomson One Banker rankings and supplement it with listings in Prime IPO database. The top ten investment banks (lead managers) in terms of market share are considered reputed; others are not. The rankings are updated annually.

Anchor reputation: Following Seth et al., (2019) we measure the reputation of an anchor investor by its market share just as the reputation of an underwriter is measured by its market share⁶. We set the reputation score equal to the market share. By design, the total of reputation scores of all anchor investors equals 100. These reputation scores are then translated into an IPO-reputation score by summing up the scores of each anchor investor backing the IPO. All IPOs with reputation scores above the median are classified as backed by reputed anchors.

Subscription (x): We measure investor subscription in three categories i.e. retail, non-institutional and Qualified Institutional Buyer (QIBs).

Investment to Assets: Capital expenditure expressed as a percentage of the beginning of the period total assets

⁶ Market share is simply the ratio of capital committed by an anchor investor in a year across all IPOs and the total capital committed by all anchor investors to all IPOs in a year. We posit that issuers would approach only those anchor investors whose investment sends a positive signal to other investors. Thus, the market share, as defined above, is a measure of reputation.

Investment to PPE: Capital expenditure expressed as a percentage of the beginning of the period property, plant and equipment

Profitability: EBITDA as the percentage of the beginning year of period total assets

Tangibility: Ratio of fixed assets and total assets

Leverage: Book value of long-term debt and short-term debt scaled by total assets

Firm Size: Natural log of total assets

3.4.Descriptive Statistics

Panel A of Table 1 reports the descriptive statistics of our sample of IPOs. It includes the mean, median, and the standard deviation of key variables used in our analysis. It displays the differences in means between anchor-backed and non-anchor-backed IPOs. The difference in firm-level variables such as the market-to-book ratio, profitability (return on assets, ROA), firm size, institutional shareholding, cash flow assets and sales growth rate are statistically significant. Our univariate analysis suggests that on average, anchor-backed firms have more growth opportunities than non-anchor-backed firms. They are more profitable, larger and have a higher percentage of institutional shareholding. Anchor-backed IPOs have higher cash flow as compared to non-anchor-backed firms. However, they have lower growth in sales as compared to non-anchor-backed firms.

Panel B of Table 1 presents the pairwise correlation among the key variables used in our analysis. We set the anchor dummy equal to 1 if the IPO is anchor-backed and zero, otherwise. We find that the anchor dummy is positively correlated with growth opportunities (M/B ratio), profitability, institutional shareholding and cash flow to assets ratio. We also find that the anchor dummy is negatively correlated with sales growth rate and insignificantly correlated with tangibility, investment, and leverage.

4. Empirical Results

In this section, we present the main results of the paper.

4.1. Empirical Strategy

We use OLS regressions to model pre-market demand, underwriter compensation and while we use propensity score matching and DiD estimation to model issue discount, investment and financial leverage after the IPO. We employ Propensity Score Matching (PSM) to match the treated with the untreated group. We define anchor-backing as the treatment, anchor-backed IPOs as the treated group and the non-anchor-backed IPOs as the untreated. The matching is done using a function of controls such as market-to-book ratio, tangibility, profitability, firm size, institutional ownership, and cash flow to assets.

Table 2 presents a description of the notation we use for exposition. It is possible to match the treated and the control group during the pre and post-AI period for Group 12, Group 21 and Group 22 but not for Group11 because we do not have a sample of AI-backed IPOs in the Pre-AI period. Therefore, we use PSM to identify the sample of AI-backed IPO in the pre-AI period by using a variety of firm-level variables. We employ four methods to match: nearest neighbor with and without replacement, nearest neighbor with replacement and caliper equal to 0.21 and Mahalanobis matching. However, we use the nearest neighbor matching without replacement in our analysis. We identify a sample of 43 firms from the Pre-AI period (Group11) and match them with AI-backed IPOs from the Post-AI period (Group12).

Once the matching is done, the entire group of AI-backed IPO and the NAI firms from the pre-AI period forms the treated group. The remaining NAI firms (from the pre-AI period) and from the post-AI period forms the untreated group. In particular, we have a sample of 197 treated firms and 591 control firms. We then match NAI firms from the pre-AI period with NAI firms

from the post-AI period. Thereafter, we employ difference-in-difference (DiD) estimation to identify the effect of anchor investment on anchor-backed IPOs. Specifically, we test whether anchor-backed IPO firms in general, and growth and financially constrained firms in particular, were able to reduce underwriter compensation, issue discount and leverage and increase investments following the regulation. Empirically, we estimate the following DiD equation:

$$Outcome_{it} = \beta_0 + \beta_1 * Post_t * Treated_{it} + \beta_2 * Treated_{it} + X_{it}\theta + \gamma_j + \mu_t + \varepsilon_{it} \quad (1)$$

Where $Post_t$ is an indicator variable to equal to one for fiscal years ending after 2009. $Treated$ is an indicator variable equal to one if the IPO is anchor-backed and zero, otherwise. X_{it} is the set of controls. The coefficient of interest is β_1 of $Treated_t * Post_t$, which is the product of $Treated$ dummy and $Post$ dummy. We also include year and industry fixed effects. We have not included $Post_t$ as a separate variable in the regression as the year fixed effects account for any time-varying changes in the outcome variable. The strength of β_1 identifies the differential effect of the regulation after controlling for a variety of firm-level characteristics and overall time trends.

We broadly segregate our analysis into pre-market and after-market effects. These are discussed below.

4.2. Pre-and Immediate After-Market Effects

4.2.1. Pre-Market Demand

Prior papers on IPOs in the U.S. have used price update as a proxy for pre-market demand because issuers set the final offer price after gauging investor demand during book building (e.g. Aggarwal et al., 2002). They do so because actual subscriptions are not observable. Since book building in India takes place online, it is possible for an investor to observe subscription patterns when the book is being built. Thus, we focus on actual subscriptions rather than update. Seth et

al., (2019) show that anchor backed IPOs take longer to go public and are less likely to be priced at either the upper end or above the mid-point of the price band. In this sub section we examine how other investors react to a firm's decision to recruit anchor investors. This is particularly important because aggregate demand information is one of the principal problems faced by issuers and investment banks. If anchor investors convey credible signal of firm quality, we would expect anchor backed IPOs to have higher subscription levels. On the flip side, the ability of anchor investors to credibly certify a firm's quality is blunted by the short lock-up expiration period⁷ and the underwriter's discretion in allocating shares to anchor investors in hot IPOs.

Table 3 reports the results of Ordinary Least Squares regressions of subscriptions in the institutional and retail investor categories. In column 1 we model the (natural log of) subscription by qualified institutional buyers (plus 1). The independent variables include an anchor dummy that takes the value of 1 if the IPO is backed by anchor investors, the percentage difference between the actual offer price and the mid-point of the filing range (update), natural log of issue proceeds, a lead manager reputation dummy, which is 1 if the lead manager is among the top ten in terms of market share, and the percent change in Nifty index between the filing date and the offer date.

When the pre-market interest for an IPO (as measured by update) is high, we expect the eventual subscription to be higher because issuers set the final offer price after gauging the demand during book building. Smaller issues have ex-ante greater uncertainty than larger issues (Beatty and Ritter, 1986) because of which subscription in smaller issues might be subdued. Changes in the market during the waiting period might affect subscriptions just as they affect the number of

⁷ Lock Ups expire 30 days from the allotment date, which is usually about 2-3 weeks from the listing date

shares offered (Hanley, 1993). We also include time and industry dummies (coefficients not reported).

We find that anchor backed IPOs have lower institutional subscription. The coefficient of IPO proceeds is positive and significant suggesting that institutional investors prefer to invest in larger IPOs. We can infer from these results that anchor investors are recruited when other institutional investors are not likely to invest thereby filling a void.

In column 2 we model the (natural log of) subscription by retail investors plus 1. The independent variables include all of the above. In addition, we consider subscription by qualified institutional buyers (plus 1). Clarke et al. (2016) show the presence of sequential learning among IPO investors in India. They argue that transparency in book building helps alleviate the winner's curse problem for retail investors and that they follow the institutional investors' subscription patterns when bidding for shares.

The results show that 1) anchor investors have insignificant impact on retail investors' subscription but other institutional investors indeed have a significant impact 2) retail investors' demand for an IPO decreases when the difference between the final offer price and the mid-point of the price range increases consistent with the notion that these investors prefer IPOs that are moderately priced and 3) retail investors prefer smaller IPOs where they have a greater chance of allocations.

4.2.2. Underwriter Compensation

Under the Indian securities law, a firm-commitment underwriting contract is mandatory for the portion of shares offered to the public. Sherman and Titman (2002) present a model in which underwriter selects a group of investors along with a pricing and allocation mechanism in a way that maximizes the information generated during the process of going public at a minimum

cost. One of the potential benefits of the anchor investors program is that it reduces the risk of devolvement for underwriters. The “reduction in underwriting risk” hypothesis suggests that by pre-selling a large fraction of shares to anchor investors, underwriters can avoid the risk of under-subscription. Since the process results in a reduction in underwriting risk we expect the underwriter compensation to reduce when anchor investors are involved even for riskier firms. To explore this issue, we run an ordinary least squares regression with the natural log of underwriter fees as the dependent variable. The control variables include the inverse of the natural log of offer size, lead manager reputation, venture capital backing, firm age and firm size that are known to affect underwriter compensation (Eckbo et al. 2007 and papers cited therein). We include (but do not report) year dummies (as underwriter compensation varies through time) and industry dummies. We estimate the following models:

$$\text{Natural Log of underwriter compensation} = \beta_0 + \beta_1 (\text{anchor dummy or reputed anchor dummy}) + \beta_{2-7} (\text{control variables}) + \text{Time Fixed Effects} + \text{Industry Fixed Effects} + \varepsilon$$

(2)

Table 4 reports the results of the regressions. In Regression 1, the independent variable of interest is the anchor dummy. The coefficient of anchor dummy is negative in line with the expectation but insignificant. In Regression 2 we replace anchor dummy with a reputed anchor dummy. Since reputed anchors contribute greater amounts of capital compared to other types of investors, we would expect the underwriter fees to fall more steeply when an IPO is backed by reputed anchors. The coefficient of reputed anchor dummy is significantly negative suggesting that underwriter compensation indeed reduces when reputed anchor investors are involved. The underwriter compensation falls by 7%. The coefficient of inverse of the natural log of proceeds is positive, which suggests that bigger IPOs attract lower fees. This is consistent with the view that there are

economies of scale in going public. The coefficient of lead manager reputation is positive, which supports the notion that reputed underwriters are likely to charge higher fees because of their distribution capabilities. In summary, our analysis provides evidence to support the fact that anchor investors could reduce the cost of going public.

4.2.3. Issue Discount

The literature has documented widespread underpricing (or issue discount or initial returns) of IPOs (Ljungqvist (2007) and Ritter and Welch (2002) provide excellent reviews of the theory and evidence on IPO underpricing and allocation). Underpricing is the most important indirect flotation costs in a security offering. There are several ways to measure underpricing of security issues. The offer price relative to the closing price on the offer date is generally termed the underpricing level. Seth et al., (2019) document a negative, causal relation between allocation to anchor investors and underpricing. In this sub-section we examine whether anchor backing reduces the issue discount during the period of our study. We estimate the following DID equation with underpricing as the dependent variable:

$$\text{Issue Discount} = \beta_0 + \beta_1 (\text{post regulation dummy} * \text{anchor dummy}) + \beta_2 (\text{anchor dummy}) + \beta_j (\text{control variables}) + \text{Time Fixed Effects} + \text{Industry Fixed effects} + \varepsilon \quad (3)$$

Treated is an indicator variable equal to 1 if the IPO is backed by anchor investors and zero, otherwise. *Post* is an indicator variable set equal to 1 for fiscal years after 2009. *Treated * Post* is the product of the two dummy variables. The regression includes industry and time fixed effects. The variable of interest is the interaction dummy. The results of our regression are presented in Table 5. We find that anchor backed IPOs have 26% lower underpricing. This result reinforces the notion that anchor investors reduce the cost of going public. This may be particularly valuable

for young firms facing asymmetric information. The sign of the coefficient on firm size (natural log of total assets) is negative but insignificant. That is, bigger firms are less underpriced (and smaller firms are more highly underpriced). Likewise, firms with less tangible assets are more highly underpriced. This result is consistent with Rock (1986). And firms with more growth opportunities (market/book ratio) and institutional investment are more highly underpriced.

Overall, our results in the preceding sub-sections suggest that, in the pre-and the immediate after-market, anchor investors reduce security issuance frictions by acting as substitutes for other types of institutional investors and reducing the cost of going public, which includes underwriter compensation and the degree of underpricing.

4.3. After Market Effects: Investment and Financial Leverage

In this sub-section, we investigate whether anchor investors have a significant impact on investment and financial leverage of IPO firms.

4.3.1 Effect on Investments

In the previous sub-section, we showed that anchor investment in IPOs results in a reduction in flotation cost. In particular, the underwriter compensation and issue discount reduce when the IPO attracts investment from anchor investors. In other words, anchor investors reduce the cost of issuing equity. This may result in higher investment in projects. To test this hypothesis, we rely on the difference-in-difference framework outlined earlier.

We estimate the following DID equation:

$$\text{Investment/Total Assets} = \beta_0 + \beta_1 (\text{post regulation dummy} * \text{anchor dummy}) + \beta_2 (\text{anchor dummy}) + \beta_j (\text{control variables}) + \text{Time Fixed Effects} + \text{Industry Fixed effects} + \varepsilon$$

(4)

Table 6 presents the results of our difference-in-difference estimation. Our primary measures of investment are capital expenditure scaled by the beginning period total assets and capital expenditure scaled by the beginning of the period property, plant, and equipment. Column 1 reports the results with capital expenditure scaled by the beginning period of total assets as the dependent variable. The coefficient β_1 of $Treated_t * Post_t$ is positive and statistically significant at the 1% level and economically large. We replicate our analysis using capital expenditure scaled by the beginning period of property, plant and equipment as the dependent variable in column 2. The coefficient β_1 of $Treated_t * Post_t$ is also positive and statistically significant at 1%. It is also important to note that the coefficient β_1 is economically large suggesting a significant increase in investments.

4.3.2 Effect on Financial Leverage

Changes in the issuance costs can impact a firm's capital structure (Fischer, Heinkel, and Zechner; 1989) too. We expect anchor backed firms to experience a reduction in financial leverage due to their ability to maximize proceeds and to lower the cost of issuing equity⁸.

We estimate the following DID equation

$$\text{Leverage} = \beta_0 + \beta_1 (\text{post regulation dummy} * \text{anchor dummy}) + \beta_2 (\text{anchor dummy}) + \beta_j (\text{control variables}) + \text{Time Fixed Effects} + \text{Industry Fixed effects} + \varepsilon \quad (5)$$

In Table 7 we tabulate the results of difference-in-difference estimation. Our primary measure of interest is *leverage* which is defined as the sum of the long term and short-term debt as a percentage of total assets. The coefficient β_1 of $Treated_t * Post_t$ in column 1 is negative and statistically significant at 5% level. In terms of economic significance, firms experience a

⁸ We explore this in a subsequent section

reduction of about 12.3% in financial leverage when the IPO is backed by anchor investors. This supports our hypothesis that due to reduced issuance costs and resolution of symmetric information; anchor-backed firms are able to reduce leverage.

5. Additional Analysis

In this section, we provide empirical evidence on the types of firms that benefit from the regulation i.e. introduction of anchor investors in IPOs. In particular, we investigate whether high growth and financially constrained firms benefit more from the involvement of anchor investors.

5.1 Impact on Growth Firms

In the previous sections we showed that the involvement of anchor investors results in a reduction in issuance costs, higher investment, and lower financial leverage. We investigate whether this is particularly true of high growth firms because these are the firms that are likely to raise capital in pursuit of profitable projects and hence, benefit from anchor investment. To examine this possibility, we divide our dataset into two and classify firms as high and low M/B by setting the cut off at the median value of the ratio of market value of equity and the book value of equity (*Market to Book*), a commonly used proxy for growth opportunities.

To test this hypothesis, we rely on difference-in-difference estimation. We estimate the following equations:

$$\text{Investment/Total Assets} = \beta_0 + \beta_1 (\text{post regulation dummy} * \text{anchor dummy}) + \beta_2 (\text{anchor dummy}) + \beta_j (\text{control variables}) + \text{Time Fixed Effects} + \text{Industry Fixed effects} + \varepsilon \quad (6)$$

$$\text{Investment/PPE} = \beta_0 + \beta_1 (\text{post regulation dummy} * \text{anchor dummy}) + \beta_2 (\text{anchor dummy}) + \beta_j (\text{control variables}) + \text{Time Fixed Effects} + \text{Industry Fixed effects} + \varepsilon \quad (7)$$

Table 8 presents the results of DiD estimation. Columns 1 and 2 present the results for firms with high market-to-book ratio while Columns 3 and 4 present the results for low M/B firms. In columns 1 and 3, the dependent variable *Investment/TA* is defined as the capital expenditure as the percentage of the beginning of period total assets (i.e. $100 * Capex_{t-1} / Total\ Assets_t$). In Columns 2 and 4, the dependent variable is *Investment/PPE* defined as the ratio of capital expenditure and the beginning of the period property, plant and equipment (i.e. $100 * Capex_{t-1} / PPE_t$). *Treated* is an indicator variable equal to one if IPO is backed by an anchor investor and zero, otherwise. *Post* is an indicator variable to equal to one for fiscal years ending after 2009. *Treated*Post* is defined as the product of Treated dummy and Post dummy.

We find that the increase in investment is observed only among firms with a high *Market-to-Book* ratio. The coefficients (β_1) of *Treated_t*Post_t* in columns 1 and 2 are positive and statistically significant at the 5% level and economically large. This supports the hypothesis that enlisting anchor investors results in higher investment only among *High M/B* firms. We do not find such an evidence for *Low M/B* firms (see columns 3 and 4). Our results are consistent with the view that firms with more growth opportunities may be able to pursue attractive growth opportunities that might otherwise have been difficult to implement when the IPO is backed by anchor investors. Next, we replicate the analysis using *leverage* as the dependent variable as follows.

$$\text{Leverage} = \beta_0 + \beta_1 (\text{post regulation dummy} * \text{anchor dummy}) + \beta_2 (\text{anchor dummy}) + \beta_j (\text{control variables}) + \text{Time Fixed Effects} + \text{Industry Fixed effects} + \varepsilon$$

(8)

In Table 9, the coefficient β_1 of *Treated_t*Post_t* in column 1 is negative and statistically significant at the 1% level, indicating that high growth firms are able to reduce leverage when

backed by anchor investors. This result is applicable only for *High M/B* firms while we do not find such an evidence for *Low M/B* firms as shown in column 2.

5.2 Impact on Financially Constrained Firms

We posit that the regulation allows financially constrained firms to access capital markets. In the absence of anchor investors such firms may have been rationed out of capital markets. To measure financial constraints, we construct the Size-Age (SA) index suggested by Hadlock and Pierce (2010). They show that size and age have meaningful predictive power in determining the level of financial constraints. Smaller and younger firms exhibit significantly more uncertainty regarding their financing ability. We employ the coefficient estimates from model (2) of Table 6 in Hadlock and Pierce (2010), which utilizes firm size, firm size squared, and firm age, to produce the SA index to proxy for the level of financial constraints. We estimate the following regression: $-0.737\text{Size} + 0.043 \text{Size}^2 - 0.040\text{Age}$, where Size is the log of inflation-adjusted (to 2004) total assets and age is the number of years the firm is listed. To calculate the index values, we follow Hadlock and Pearce (2010) and redefine size-age capping in our context and follow Farre-Mensa and Ljungqvist (2013) to sort firms into terciles based on their index values. Firms in the top tercile are coded as constrained and those in the bottom tercile are coded as unconstrained firms. We estimate the following equation:

$$\text{Equity Proceeds} = \beta_0 + \beta_1 (\text{post regulation dummy} * \text{anchor dummy}) + \beta_2 (\text{anchor dummy}) + \beta_j (\text{control variables}) + \text{Time Fixed Effects} + \text{Industry Fixed effects} + \varepsilon \quad (9)$$

Table 10 presents the results of difference-in-difference estimation. Column 1 suggests that the regulation has a significant impact on severely constrained firms. After the regulation, constrained firms experience an increase of approximately 26 % in their offer proceeds, which is not observed in the case of unconstrained firms.

In Table 11 we provide evidence that the regulation allows constrained firms to reduce financial leverage as well. We replicate our analysis using *leverage* as the dependent variable. The coefficient β_1 of $Treated_t * Post_t$ in column 1 is negative and statistically significant at the 5% level. This result holds only for constrained firms while we do not observe a similar effect among unconstrained firms (column 2).

5.3. Long-term operating performance

In this section, we examine the impact of anchor investors on long term operating performance of firms. Given the information content of anchor investment as well as the monitoring role of anchor investors, we expect anchor-backed firms to have better long-term operating performance. We estimate the following DID regression equation:

$$OP = b_0 + b_1 \text{Date Dummy} + b_2 \text{Anchor Dummy} + b_3 \text{DID} + \varepsilon \quad (10)$$

Where OP is the operating performance, Date Dummy is a dummy variable for IPOs launched before/after 2009, and Anchor Dummy is an indicator variable set equal to 1 if the IPO is backed by anchor investors and DID is the product of Date Dummy and Anchor Dummy

In Table 12 we report the results. The dependent variable, *operating performance*, is defined as the *earnings before interest, taxes and depreciation* scaled by beginning of the year *total assets*. The variable of interest is DiD. The coefficient on DID is positive and significant at 1% level of significance. In terms of economic significance, anchor backed firms experience an improvement of about 7.3% in their post-issue operating performance after the regulation. As a robustness test, we replicate the analysis using operating performance defined as the *earnings before interest, taxes and depreciation* scaled by beginning year of *Sales*. The result in Table 13 suggests that the coefficient of DID is not statistically significant but is economically significant. These results broadly support our hypothesis that anchor investors can create value through

significant improvement in post-issue operating performance of investee firms.

6. Conclusion

Prior research suggests that allocation of shares to anchor investors in an initial public offering (IPO) has a significant impact on the valuation and underpricing of IPOs primarily due to the mitigation of *ex-ante* information asymmetry. We report additional evidence on the monitoring role of anchor investors. We document a causal negative relation between anchor backing and 1) the cost of going public and 2) financial leverage but a causal positive relation with 1) investments and 2) equity proceeds. We also find that anchor investment could be particularly valuable for high growth and capital constrained firms. Anchor backed firms exhibit better operating performance after the IPO. Overall, we conclude that discretionary allotment of shares to anchor investors has a positive impact on the performance of firms.

Appendix A: Construction of Variables

Variable	Variable definitions	Data Sources
Underpricing	Percentage return from the offer price to the first day closing price	Prime IPO Database
Anchor Reputation	Market share held by anchor investors	Prime IPO Database
Anchor Dummy	Indicator variable equal to one if IPO is backed by an anchor investor and zero, otherwise	Prime IPO Database
IPO Price Band	Book building price range of the IPO	Prime IPO Database
ln (Underwriter Fees)	Natural log of underwriter fees charged by the underwriter	Prime IPO Database
Lead Manager Reputation	Indicator variable equal to one if the IPO is lead managed by one of the top 10 lead managers in terms of market share	Thomson ONE, Prime IPO Database
Update	$((\text{Offer price} - \text{midpoint of the filing range}) / (\text{midpoint of the filing range})) \times 100\%$	CMIE PROWESS Database, Author calculation
Offer Proceeds	Natural log of IPO proceeds	Prime IPO Database

ln (QIB Subscription+1)	Natural log of subscription by institutional investors	Company Website, Prime IPO Database, Chittorgarh.com
ln (Retail Subscription+1)	Natural log of subscription by retail investors	Company Website, Prime IPO Database, Chittorgarh.com
Venture Capital Affiliation	Indicator variable equal to one if IPO is backed by venture capitalist and zero, otherwise	CMIE PROWESS Database
Nifty Index Returns	Percentage change in S&P Nifty Index between filling and offer dates	National Stock Exchange
Investment to Assets	Capital expenditure expressed as a percentage of the beginning of the period total assets	CMIE PROWESS Database
Investment to PPE	Capital expenditure expressed as a percentage of the beginning of the period property, plant, and equipment	CMIE PROWESS Database
Profitability	EBITDA as the percentage of the beginning year of period total assets	CMIE PROWESS Database
M/B ratio	Market value of equity divided by book value of equity	CMIE PROWESS Database

Tangibility	Ratio of fixed assets and total assets	CMIE PROWESS Database
Leverage	Book value of long-term debt and short-term debt scaled by total assets	CMIE PROWESS Database
Firm Size	Natural log of total assets	CMIE PROWESS Database
Institutional Shareholding	Percentage of common stock held by institutional investors	CMIE PROWESS Database
Cash Flow to Assets	Cash flow from operations expressed as a percentage of the beginning of the year total assets	CMIE PROWESS Database
Sales Growth Rate (%)	Percentage change in sales over the past year	CMIE PROWESS Database

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Table 1: Sample Characteristics**Panel A: Descriptive statistics**

This table reports the summary statistics of the key variables and univariate tests of difference of means between Anchor-backed and Non-Anchor-backed IPOs. Our sample consists of a total of 788 IPOs out of which 154 are anchor-backed and 634 are non-anchor-backed IPOs. All the variables are minorized at the 1% tail. Other variables include Profitability (ratio of EBITDA to total assets), market to book ratio (M/B), Leverage (book value of long-term debt scaled by total assets), tangibility (ratio of fixed assets to total assets), firm size (ln total assets), Investment (capital expenditure as the percentage of the beginning year of period total assets), Cash flow from operations scaled by total assets, percentage of shares held by institutional investors and sales growth rate. The significance of the differences in means is based on the Student t-test. The asterisk superscripts ***, **, and * represent significance at the 1%, 5%, and 10% levels, respectively.

Variables	Full sample			Anchor-backed IPO	Non- Anchor-backed IPO	Difference t-statistic
	Mean	Std. Dev.	Median			
M/B ratio	2.59	2.23	1.92	4.02	2.25	5.05***
Leverage	0.64	0.10	0.63	0.63	0.64	-0.31
Tangibility	0.18	0.18	0.12	0.19	0.18	0.60
Profitability	0.14	0.12	0.13	0.16	0.13	4.35***
Firm size	7.22	2.09	7.02	9.32	6.72	15.9***
Investment	0.03	27.19	0.0	0.02	0.04	-0.01
Institutional shareholding %	8.52	9.99	5.88	16.09	6.90	5.8***
Cash flow to Assets	-0.08	0.40	-0.01	0.06	-0.11	5.6***
Sales growth rate	29.60	45.20	19.70	23.12	31.28	-1.84*

Panel B: Correlation Matrix of Key Variables

This table presents the correlation between the key variables used in our analysis. *, **, *** represent significance at 10%, 5%, and 1% levels.

Variables	Anchor dummy	Investment	Leverage	M/B ratio	Tangibility	Profitability	Firm size	Institutional Shareholdings (%)	Cash flow to Assets	Sales growth rate (%)
Anchor dummy	1.000									
Investment	0.000	1.000								
Leverage	-0.022	0.069	1.000							
M/B ratio	0.319***	-0.134*	0.060	1.000						
Tangibility	0.022	-0.045	0.105	0.137*	1.000					
Profitability	0.143***	-0.034	0.097	0.191***	0.152***	1.000				
Firm Size	0.493***	0.003	0.333***	0.273***	0.021	0.080**	1.000			
Institutional Shareholding (%)	0.349***	-0.053	-0.022	0.270***	0.188***	0.091	0.452***	1.000		
Cash flow to Assets	0.178***	-0.005	0.032	0.186***	0.204***	0.282***	0.223***	0.129*	1.000	
Sales growth rate (%)	-0.073*	0.024	0.044	-0.042	-0.017	-0.081**	-0.020	-0.056	0.010	1.000

Table 2: Propensity Score Matching notation

This table presents the notation we follow in our PSM analysis.

	Pre-AI Period	Post-AI Period
AI-backed IPO	Group 11	Group 12
NAI-backed IPO	Group 21	Group 22

Table 3: Pre-Market Demand

This table reports the results of OLS regressions of subscriptions in the institutional and retail investor categories. The dependent variable in Regressions 1 and 2 are the natural logs of subscription by qualified institutional buyers plus 1 and retail investors plus 1, respectively. The independent variables include an anchor dummy that takes the value of 1 if the IPO is backed by anchor investors, the percentage difference between the actual offer price and the mid-point of the filing range (Update), natural log of issue proceeds, a LM reputation dummy, which is 1 if the lead manager is among the top ten underwriters in terms of markets share in that year, and the percent change in S&P CNX Nifty index between the filing date and the offer date, and subscription by qualified institutional buyers plus 1. The regressions include time and industry fixed effects. The t statistics are in parenthesis. The superscripts ***, **, and * denote significance at the 1%, 5% and 10% respectively.

	ln (QIB Sub+1)	ln (Retail Sub+1)
Anchor Dummy	-0.181** (-1.947)	0.025 (0.325)
Update	-0.019 (-0.297)	-0.119** (-2.257)
ln (Proceeds)	0.268*** (4.028)	-0.317*** (-5.589)
LM Reputation	0.031 (0.351)	-0.022 (-0.315)
% Change in the NIFTY index between filing and opening dates	-0.006 (-0.037)	-0.072 (-1.341)
ln (QIB Sub +1)		0.656*** (11.474)
Intercept	-1.245 (-1.465)	3.112 (6.482)
Adj. R ²	12.8%	41.2%

Table 4: Underwriter Compensation

This table reports the results of OLS regressions of underwriter compensation. The dependent variables in Regressions 1 and 2 are the natural logs of underwriter fees. The independent variables include an anchor dummy that takes the value of 1 if the IPO is backed by anchor investors, a reputed anchor dummy that takes the value of 1 if the IPO is backed by reputed anchor investors, inverse of natural log of issue proceeds, a LM reputation dummy, which is 1 if the lead manager is among the top ten in terms of markets share, VC backed, a dummy that takes on the value of 1 if the issue is backed by venture capitalists, the natural log of firm age plus 1, and the natural log of sales. The regressions include time and industry fixed effects. The t statistics are in parenthesis. The superscripts ***, **, and * denote significance at the 1%, 5% and 10% respectively.

	(1)	(2)
Anchor dummy	-0.095 (-1.407)	
Reputed Anchor dummy		-0.851*** (-2.659)
Inverse natural log Proceeds	-0.165** (-2.428)	0.742** (2.320)
LM reputation	0.156** (2.301)	0.167** (2.510)
VC Backed?	0.096 (1.422)	0.126* (1.894)
Ln (Firm Age+1)	-0.018 (-0.261)	-0.007 (-0.103)
Ln (Sales)	0.022 (0.324)	0.019 (0.284)
Intercept	8.590*** (10.705)	6.996*** (16.336)
Observations	788	788
Adj. R-squared	0.042	0.054

Table 5: Effect on Issue Discount

This table reports the results of DID regression. The dependent variable *Issue Discount* is defined as the percentage difference between the offer price and the closing price on the listing day. *Treated* is an indicator variable equal to one if IPO is backed by anchor investors and zero, otherwise. *Post* is an indicator variable equal to one for fiscal years after 2009. *Treated*Post* is defined as the product of Treated dummy and Post dummy. All explanatory variables are defined in Appendix A. The regression includes time and industry fixed effects. The asterisk superscripts *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Issue Discount
Treated*Post	-26.120* (-1.662)
Anchor dummy	15.831 (1.184)
ln (Total assets)	-2.204 (-1.427)
M/B ratio	2.428** (2.239)
Profitability	37.399 (1.240)
Tangibility	-46.965** (-2.578)
Institutional Ownership	0.488* (1.903)
Intercept	9.461 (0.335)
Observations	788
Adj. R-squared	0.322

Table 6: Effect on Investment

This table reports the results of DID regression. In column 1, the dependent variable *Investment/TA* is defined as the ratio of capital expenditure and the beginning of the period total assets (i.e. $100 \times \text{Capex}_{t-1} / \text{Total Assets}_t$). In Column 2, the dependent variable *Investment/PPE* is defined as the ratio of capital expenditure and the beginning of period property, plant and equipment (i.e. $100 \times \text{Capex}_{t-1} / \text{PPE}_t$). *Treated* is an indicator variable equal to one if the IPO is backed by an anchor investor and zero, otherwise. *Post* is an indicator variable to equal to one for fiscal years after 2009. *Treated*Post* is defined as the product of Treated dummy and Post dummy. All explanatory variables are defined in Appendix A. The regressions include the year and industry fixed effects. The asterisk superscripts *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Investment To Assets	Investment To PPE
Treated*Post	40.994*** (2.962)	249.485*** (8.125)
Anchor dummy	-41.541*** (-3.188)	-242.315*** (-8.356)
Ln (Total assets)	0.712 (0.594)	2.084 (0.782)
Tangibility	27.190*** (2.646)	-17.389 (-0.767)
Profitability	-11.568 (-0.486)	31.482 (0.601)
Sales growth rate	0.066* (1.948)	0.254*** (3.240)
Cash flow/TA	-0.870 (-0.093)	-25.932 (-1.201)
Institutional Ownership	-0.330* (-1.776)	-0.484 (-1.188)
Intercept	34.991 (1.386)	239.590*** (4.318)
Observations	788	788
Adj. R-squared	0.290	0.530

Table 7: Effect on Leverage

This table reports the results of DID regression. The dependent variable *leverage* is measured as the sum of the long term plus short-term debt as a percentage of total assets. *Treated* is an indicator variable equal to one if the IPO is backed by anchor investors and zero, otherwise. *Post* is an indicator variable to equal to one for fiscal years ending after 2009. *Treated*Post* is defined as the product of Treated dummy and Post dummy. All explanatory variables are defined in Appendix A. The regression includes the year and industry fixed effects. The asterisk superscripts *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Leverage
Treated*Post	-0.123** (-2.238)
AB dummy	0.007 (0.153)
Ln (Total assets)	0.034*** (7.593)
Tangibility	0.022 (0.473)
M/B ratio	0.000 (0.005)
Profitability	0.118 (1.184)
Institutional Ownership	-0.000 (-0.458)
Intercept	0.473*** (4.817)
Observations	788
Adj. R-squared	0.458

Table 8: Effect on Investment in High Growth Firms

This table reports the results of DID regressions. Columns 1 and 2 report result for high market-to-book ratio firms while Columns 3 and 4 for low market-to-book ratio firms. The classification is based on the cut off at the median. In columns 1 and 3, the dependent variable *Investment/TA* is defined as the ratio of capital expenditure and the beginning of the period total assets (i.e. $100 \times \text{Capex}_{t-t+1} / \text{Total Assets}_t$). In Columns 2 and 4, the dependent variable *Investment/PPE* is defined as the ratio of capital expenditure and the beginning of period property, plant and equipment (i.e. $100 \times \text{Capex}_{t-t+1} / \text{PPE}_t$). *Treated* is an indicator variable equal to one if the IPO is backed by anchor investors and zero, otherwise. *Post* is an indicator variable to equal to one for fiscal years ending after 2009. *Treated*Post* is defined as the product of Treated dummy and Post dummy. All explanatory variables are defined in Appendix A. The regressions include the year and industry fixed effects. The asterisk superscripts *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	High M/B		Low M/B	
	Investment To Assets	Investment To PPE	Investment To Assets	Investment To PPE
Treated*Post	43.301** (2.224)	200.980*** (6.143)	1.737 (0.083)	-5.653 (-0.638)
Anchor dummy	-49.180*** (-2.694)	-192.207*** (-6.261)	5.437 (0.400)	5.066 (0.881)
Ln (Total assets)	1.744 (0.965)	0.675 (0.221)	0.501 (0.312)	0.437 (0.641)
Tangibility	-3.458 (-0.181)	-54.965* (-1.693)	64.617*** (3.708)	11.786 (1.590)
Cash flow/TA	41.406** (2.516)	-57.256** (-2.046)	17.684 (1.183)	2.333 (0.338)
Profitability	-59.987 (-1.208)	58.316 (0.697)	-87.529** (-2.367)	-7.512 (-0.478)
Institutional Ownership	-0.119 (-0.440)	-0.097 (-0.214)	-0.294 (-0.539)	-0.027 (-0.118)

Intercept	-8.005	69.834	14.627	-5.454
	(-0.233)	(1.204)	(0.750)	(-0.660)
Observations	686	686	102	102
Adj. R-squared	0.295	0.522	0.472	0.374

Table 9: Effect on Leverage in High Growth Firms

This table presents the results of DID regressions. Column 1 pertains to firms with above-the-median market-to-book ratio and Column 2 pertains to firms below-the-median market-to-book ratio. The dependent variable *leverage* is measured as the sum of the long term plus short-term debt as a percentage of total assets. *Treated* is an indicator variable equal to one if IPO is backed by an anchor investor and zero, otherwise. *Post* is an indicator variable to equal to one for fiscal years after 2009. *Treated*Post* is defined as the product of *Treated* dummy and *Post* dummy. All explanatory variables are defined in Appendix A. The regressions include the year and industry fixed effects. The asterisk superscripts *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	High M/B	Low M/B
	Leverage	
Treated*Post	-0.283*** (-2.794)	-0.021 (-0.249)
AB dummy	0.164* (1.710)	-0.019 (-0.360)
Ln (Total assets)	0.021*** (3.215)	0.047*** (7.832)
Tangibility	0.025 (0.329)	0.071 (1.107)
M/B ratio	0.009** (2.055)	0.014 (0.514)
Profitability	0.021 (0.117)	-0.009 (-0.078)
Institutional Ownership	0.001 (0.711)	-0.003 (-1.305)
Intercept	0.561*** (4.934)	0.336*** (4.224)
Observations	686	102
Adj. R-squared	0.554	0.752

Table 10: Effect on Equity Issuance for Financially Constrained Firms

This table reports the results of DID estimation. Column 1 reports the results for *constrained firms* while Column 2 for *unconstrained firms*. To measure financial constrained firms, we construct Size-Age (SA) index suggested by Hadlock and Pierce (2010) as $-0.737\text{Size} + 0.043\text{Size}^2 - 0.040\text{Age}$, where Size is the log of inflation-adjusted (to 2004) total assets and age is the number of years the firm is listed. We also cap Size at (the log of) 4.5 billion and Age at 37 years and then sort firms into terciles based on their index values. Firms in top tercile are *constrained firms* while firms in the bottom tercile are *unconstrained firms*. The dependent variable is the offer proceeds expressed as the percentage of the beginning of period market capitalization. *Treated* is an indicator variable equal to one if IPO is backed by anchor investors and zero, otherwise. *Post* is an indicator variable to equal to one for fiscal years ending after 2009. *Treated*Post* is defined as the product of Treated dummy and Post dummy. All explanatory variables are defined in the Appendix. The regressions include year and industry fixed effects. The asterisk superscripts *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Constrained Firms	Unconstrained Firms
	Equity Proceeds	
Treated*Post	0.264** (2.069)	-0.064 (-0.245)
Anchor dummy	-0.083 (-0.824)	-0.074 (-0.285)
Ln (Total assets)	-0.056*** (-4.137)	-0.011 (-0.403)
M/B ratio	-0.051*** (-4.784)	-0.034*** (-2.736)
Profitability	-0.330 (-1.203)	-0.381 (-0.726)
Tangibility	0.347** (2.287)	-0.110 (-0.531)
Institutional Ownership	-0.003 (-1.273)	0.002 (0.403)
Intercept	0.881*** (4.013)	0.420 (1.370)
Observations	394	394
Adj. R-squared	0.623	0.508

Table 11: Effect on Leverage for Financially Constrained Firms

This table presents the results of DID regression. Column 1 reports the results for *constrained firms* while Column 2 for *unconstrained firms*. To measure financial constrained firms, we construct Size-Age (SA) index suggested by Hadlock and Pierce (2010) as $-0.737\text{Size} + 0.043\text{Size}^2 - 0.040\text{Age}$, where Size is the log of inflation-adjusted (to 2004) total assets and age is the number of years the firm is listed. We also cap Size at (the log of) 4.5 billion and Age at 37 years and then sorting firms into terciles based on their index values. Firms in the top tercile are *constrained firms* while firms in bottom tercile are *unconstrained firms*. The dependent variable *leverage* is measured as the sum of the long term plus short term debt as a percentage of total assets. *Treated* is an indicator variable equal to one if IPO is backed by an anchor investor and zero, otherwise. *Post* is an indicator variable to equal to one for fiscal years ending after 2009. *Treated*Post* is defined as the product of Treated dummy and Post dummy. All explanatory variables are defined in Appendix A. The regressions include year and industry fixed effects. The asterisk superscripts *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Constrained Firms	Unconstrained Firms
	Leverage	
Treated*Post	-0.110** (-1.993)	-0.055 (-0.524)
Anchor dummy	0.027 (0.580)	-0.107 (-1.250)
M/B ratio	-0.012** (-2.593)	0.010* (1.889)
Ln (Total assets)	0.037*** (6.532)	0.036*** (3.607)
Tangibility	0.066 (0.963)	-0.116 (-1.319)
Profitability	0.072 (0.592)	0.449* (2.016)
Institutional Ownership	0.001 (0.609)	0.001 (0.294)
Intercept	0.502*** (5.097)	0.355*** (2.698)
Observations	394	394
Adj. R-squared	0.576	0.588

Table 12: DID Estimation of Operating Performance

This table reports the difference in operating performance between anchor-backed and non- anchor-backed initial public offerings (IPOs) using the difference-in-difference (DID) estimation procedure. The regression equation is: $OP = b_0 + b_1 \text{Date Dummy} + b_2 \text{Anchor Dummy} + b_3 \text{DID}$, where OP is operating performance defined as the *earnings before interest, taxes and depreciation* scaled by the beginning of the year *total assets*, Date Dummy is a dummy variable for IPOs launched after 2009, and Anchor Dummy and DID, which is the product of Date Dummy and Anchor Dummy. The asterisk superscripts *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	EBITDA_TA	Std. Error	T-stat	P-value
Before				
Non AI-backed IPO (C)	0.155			
AI-backed IPO (T)	0.138			
Diff (T-C)	-0.017	0.02	-0.86	0.393
After				
Non AI-backed IPO (C)	0.113			
AI-backed IPO (T)	0.169			
Diff (T-C)	0.056	0.011	5.05	0.000***
Diff-in-Diff	0.073	0.023	3.21	0.001***

Table 13: DID Estimation of Operating Performance

This table reports the difference in operating performance between anchor-backed and non- anchor-backed initial public offerings (IPOs) using the difference-in-difference (DID) estimation procedure. The regression equation is: $OP = b_0 + b_1 \text{Date Dummy} + b_2 \text{Anchor Dummy} + b_3 \text{DID}$, where OP is operating performance, defined as the *earnings before interest, taxes and depreciation* scaled by the beginning of the year Sales, Date Dummy is a dummy variable for IPOs launched after 2009, and Anchor Dummy and DID which is the product of Date Dummy and Anchor Dummy. The asterisk superscripts *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	EBITDA_ Sales	Std. Error	T-stat	P-value
Before				
Non AI-backed IPO (C)	2.559			
AI-backed IPO (T)	6.561			
Diff (T-C)	4.002	14.581	0.27	0.784
After				
Non AI-backed IPO (C)	1.718			
AI-backed IPO (T)	25.639			
Diff (T-C)	23.921	8.538	2.8	0.005***
Diff-in-Diff	19.919	16.897	1.18	0.239